

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, SQL, C++, Java, Bash

LLM & Agentic Systems

LLM agents, tool use, function calling, agent frameworks, provider routing/fallback, prompt design, evaluation, OpenAI / Groq APIs, Ollama, MCP

Retrieval & Knowledge Graphs

RAG, knowledge graphs, multi-view retrieval, RRF, cross-encoder reranking, MMR, knapsack token budgeting, Brave Search, sentence-transformers, llama-index, spaCy

Full-Stack & Backend

React, Electron, FastAPI, Node.js, Express, REST APIs, async services, integration testing, MongoDB, PostgreSQL, Redis, SQLite

Cloud & DevOps

AWS, GCP, Azure, Docker, Cloudflare Workers / Durable Objects / D1, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected and shipped a **full-stack internal product** (React, Node.js, MongoDB) used daily by the AI team to ingest, label, and centralize training data; replaced fragmented spreadsheet workflows with secure role-based REST APIs and reviewer dashboards, cutting manual operations by **100+ hours/month** and lifting throughput **60%**.
- Owned reliability and security for the **multi-tenant NHS eTriage backend** (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly submissions**, integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines; hardened auth and rate limiting, remediated **20+ VAPT findings**, and contributed to **\$340K+** in revenue.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and shipped **20+ production RESTful APIs** in Node.js + Fastify over **PostgreSQL** and integrated **LLM-powered context-aware replies** into a customer-facing chat product, iterating on prompt design and response-quality heuristics based on real usage.

Projects

AgenticSearch - Agentic LLM Pipeline for Structure from Messy Data

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built an **agentic LLM service** that ingests web documents at scale and mines structured entity tables from messy/unstructured sources via typed retrieval planning, page rerank, deterministic parsers with **LLM fallback**, output verification, and **cell-level provenance** for every extracted result.
- Hardened the service with **150+ automated tests**, provider routing/fallback across OpenAI/Groq backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality, source URLs, and ranking transparency.

KG2RAG-Enhanced - Graph-Based Knowledge Retrieval for Multi-Hop QA

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, llama-index, spaCy, PyTorch | UMass CS 685 Advanced NLP

- Extended a KG-guided RAG pipeline with **multi-view seed retrieval** for multi-hop QA on HotpotQA: sub-question decomposition, per-view passage retrieval over a knowledge graph, and fusion via Reciprocal Rank Fusion (RRF), cross-encoder reranking, and MMR.
- Replaced heuristic top-M selection with a **0-1 knapsack token-budget** formulation (value = relevance × coverage) for context compaction; ran **100-500-question batch experiments** via an Ollama + LLaMA-3 inference pipeline with token and latency logging.

AutoEval - Automation Agent with Auditable Acceptance Trail

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an **automation agent** that proposes new queries and labels, scores them against multiple reference systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective, with anti-Goodharting guardrails and **Git-backed acceptance gates** so every accepted change ships with an auditable report card and reusable template.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to open-source projects (Linkfree, Freehit, ProjectsHut). **Sonare** (Qualcomm Edge AI Hackathon 2025): offline sign ↔ speech cross-platform app shipped under hackathon time pressure.
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.